Cryptography

Elliptic-Curve Cryptography

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Agenda For Today

1. Shorter keys, same safety
Why another kind of crypto

2. A curve of points
$y^2 = x^3 + ax + b \bmod p$

3. Adding points
Chord, tangent, formulas

4. Multiplying by a number
kG and the ECDLP

5. Agreeing on a key
ECDH, X25519, HKDF

6. Encrypting and signing
ElGamal and ECDSA on a curve

7. ECC in the real world
Sizes, speed, where it lives

Today you will need



version 3.8 or newer



any one you like



Task 1 and small sums

Nothing to install. Starter files for Task 2 to Task 6: **GitHub**

Recall: Diffie-Hellman and RSA

Diffie-Hellman

both compute $g^{ab} \bmod p$

safe while the **discrete logarithm** is hard: from $g^a \bmod p$, find a

RSA

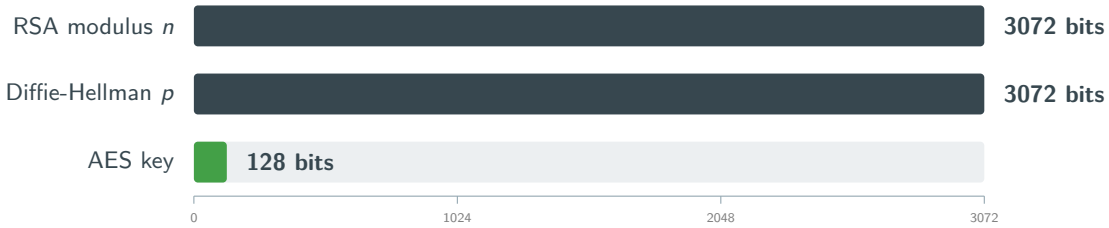
$C = P^e \bmod n$

safe while **factoring** is hard: split n into its primes

Both use a **public key** that everybody may know and a **private key** that only its owner knows.

The price: long keys

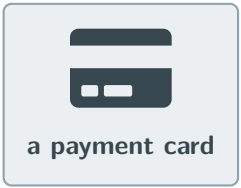
To be as hard to break as trying 2^{128} keys:



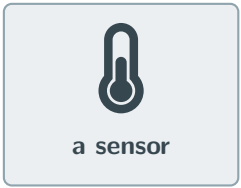
The better the attacks, the longer the keys must be.

Fast attacks on factoring and discrete logarithms force very long keys.

Where long keys hurt



a small chip,
little memory, little power



a battery that
must last years



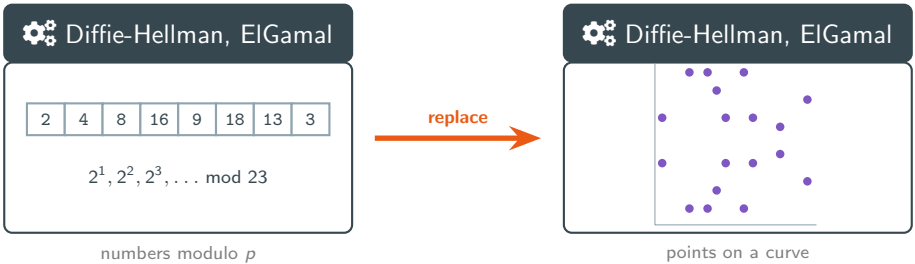
thousands of
connections a day



sent with every
HTTPS connection

The idea: a harder group

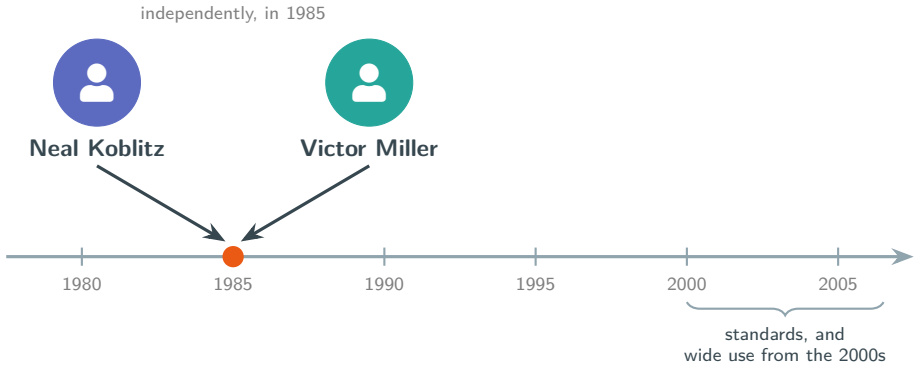
Keep Diffie-Hellman and ElGamal exactly as they are, but replace "numbers modulo p" with "points on a curve".



On a good curve, the best known attack is plain guessing. So **256 bits** give **128 bits** of security.

The same ideas on a curve: keys about 12 times shorter.

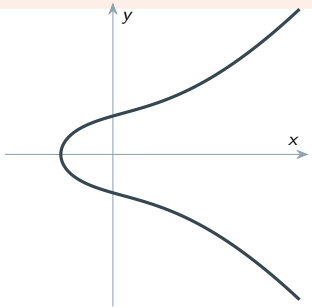
Elliptic-curve cryptography, 1985



ECC: elliptic-curve cryptography.

An elliptic curve

Elliptic curve: all points (x, y) with $y^2 = x^3 + a\,x + b$, for chosen numbers a and b.



Here: $a = 2$, $b = 2$:

$$y^2 = x^3 + 2x + 2$$

over the real numbers.

One line, the same above and below the x-axis.

Real numbers will not do

Computers need exact numbers. Over the reals, points have endless decimals:

$$x = 1, \qquad y = \sqrt{5} = 2.236067977499789696409173668731276235\cdots$$

never ends, never exact

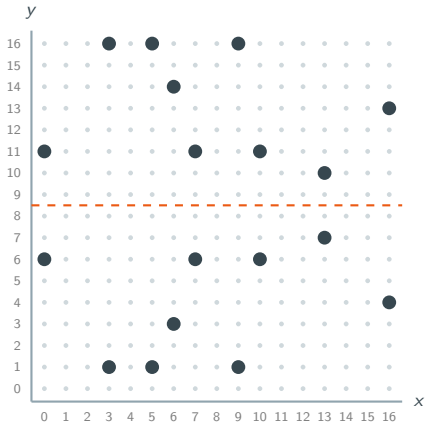
So we do everything **modulo a prime** p, as in Diffie-Hellman.

The same curve modulo 17

$$y^2 = x^3 + 2x + 2 \bmod 17$$

x and y: whole numbers from 0 to 16.

- a point of the curve
- not on the curve
- the line $y = 8.5$: (x, y) and $(x, 17 - y)$ are mirror images



Modulo p, the curve is a cloud of dots.

Is (5, 1) on the curve?

$x = 5,\ y = 1$

left side

$$y^2 = 1^2 = 1$$

right side

$$x^3 + 2x + 2 = 125 + 10 + 2 = 137$$

$$137 = 8 \times 17 + 1 \ \rightarrow\ \mathbf{1} \bmod 17$$



Both sides are 1: $(5, 1)$ is on the curve.

All the points

18 points with x and y from 0 to 16, in mirror pairs (x, y) and $(x, 17 - y)$:

(0,6) (3,1) (5,1) (6,3) (7,6) (9,1) (10,6) (13,7) (16,4)

(0,11) (3,16) (5,16) (6,14) (7,11) (9,16) (10,11) (13,10) (16,13)

+ O

the point at infinity

Plus one extra point: the **point at infinity** O.

This curve has 19 points: 18 dots and O.

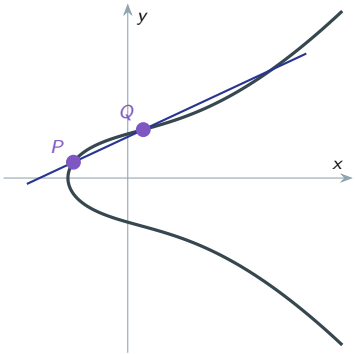
Real curves

	curve	prime p	number of points
	our toy curve	17	19
	P-256 (NIST)	$2^{256} - 2^{224} + \ldots$	about 2^{256}
	Curve25519	$2^{255} - 19$	about 2^{252} in its main group

Too many points to list, but the rules are the same.

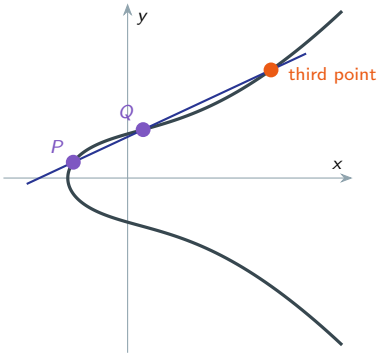
Adding two points: the idea

1. Draw a straight line through P and Q.



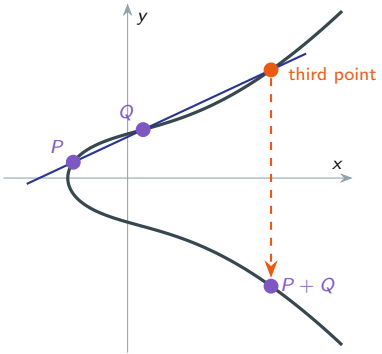
Adding two points: the idea

1. Draw a straight line through P and Q.
2. It meets the curve in exactly **one more point**.



Adding two points: the idea

1. Draw a straight line through P and Q.
2. It meets the curve in exactly **one more point**.
3. Reflect that point in the x-axis: this is $\mathbf{P} + \mathbf{Q}$.

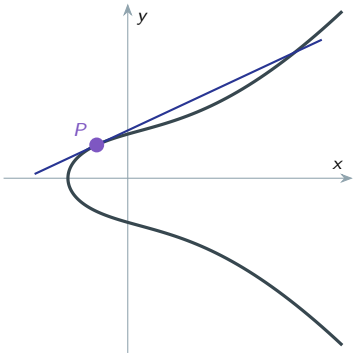


$P + Q$: draw the line, find the third point, reflect it.

Doubling a point

To add P to itself, the line through P and P is the **tangent** at P.

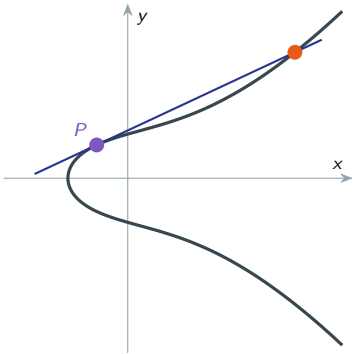
1. Draw the tangent at P.



Doubling a point

To add P to itself, the line through P and P is the **tangent** at P.

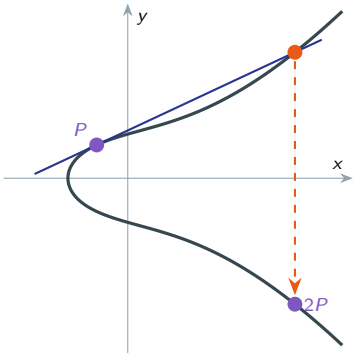
1. Draw the tangent at P.
2. Find where it meets the curve again.



Doubling a point

To add P to itself, the line through P and P is the **tangent** at P.

1. Draw the tangent at P.
2. Find where it meets the curve again.
3. Reflect: this is $P + P = \mathbf{2P}$.



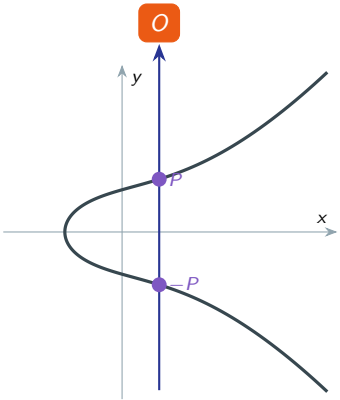
P + (-P) = O

$-P$ is the reflection of P:

$$P = (x, y), \qquad -P = (x, -y)$$

The line through P and $-P$ is **vertical**.
It never meets the curve again.

We say it meets the curve at O,
the **point at infinity**.



O works like zero: $P + O = P$ and $P + (-P) = O$.

The formulas

slope s: for $P + Q$: $s = (y_2 - y_1) / (x_2 - x_1) \mod p$

for $P + P$: $s = (3x_1^2 + a) / (2y_1) \mod p$

result:

$$x_3 = \mathbf{s}^2 - x_1 - x_2 \mod p$$

$$y_3 = \mathbf{s}(x_1 - x_3) - y_1 \mod p$$

"$/\, v \bmod p$" means multiplying by the **modular inverse** of v: the number v^{-1} with $v \cdot v^{-1} = 1 \bmod p$.

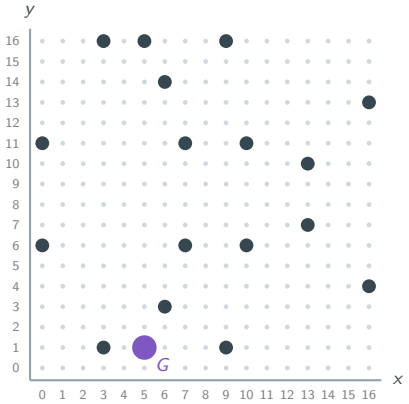
Doubling G, step 1

$G = (5, 1)$ on $y^2 = x^3 + 2x + 2 \bmod 17$, $\quad 2G = ?$

- **the slope** $s = (3 \times 5^2 + 2)/(2 \times 1) = 77/2 \bmod 17$

  $77 \bmod 17 = 9, \quad 2^{-1} \bmod 17 = 9$

  $s = 9 \times 9 = 81 \bmod 17 = \mathbf{13}$



Doubling G, step 2

$G = (5, 1)$ on $y^2 = x^3 + 2x + 2 \bmod 17$, $\quad 2G = ?$

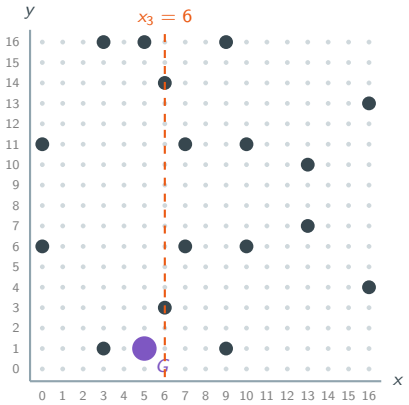
the slope $s = (3 \times 5^2 + 2)/(2 \times 1) = 77/2 \bmod 17$

$77 \bmod 17 = 9, \quad 2^{-1} \bmod 17 = 9$

$s = 9 \times 9 = 81 \bmod 17 = \mathbf{13}$

- **x_3** $x_3 = 13^2 - 5 - 5 = 159 \bmod 17$

  $x_3 = \mathbf{6}$



Doubling G, step 3

$G = (5, 1)$ on $y^2 = x^3 + 2x + 2 \bmod 17$, $\quad 2G = ?$

the slope $s = (3 \times 5^2 + 2)/(2 \times 1) = 77/2 \bmod 17$

$77 \bmod 17 = 9, \quad 2^{-1} \bmod 17 = 9$

$s = 9 \times 9 = 81 \bmod 17 = \mathbf{13}$

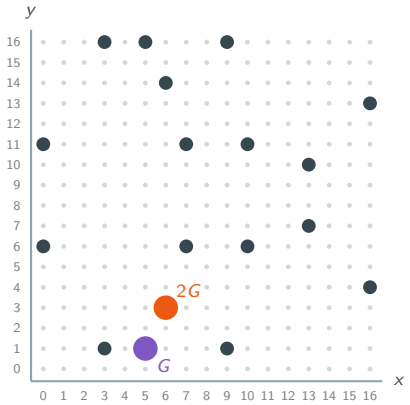
x_3 $x_3 = 13^2 - 5 - 5 = 159 \bmod 17$

$x_3 = \mathbf{6}$

y_3 $y_3 = 13 \times (5 - 6) - 1 = -14 \bmod 17$

$y_3 = \mathbf{3}$

$2G = (6, 3)$. Check: $3^2 = 9$ and $6^3 + 12 + 2 = 230 = 13 \times 17 + 9$.



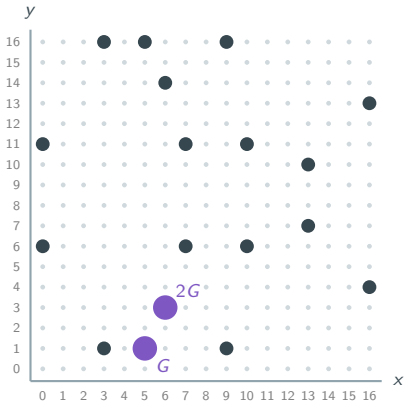
$2G = (6, 3)$ is again a point of the curve.

Task 1: G + 2G on paper

Task 1 (on paper)
$G = (5, 1)$, $2G = (6, 3)$, on $y^2 = x^3 + 2x + 2 \bmod 17$

1. Compute the slope s for **two different points**.
2. Compute x_3 and y_3.
3. Check that the result is on the curve.

$$3G = G + 2G = (\ ?\ ,\ ?\ )$$



Solution: G + 2G on paper

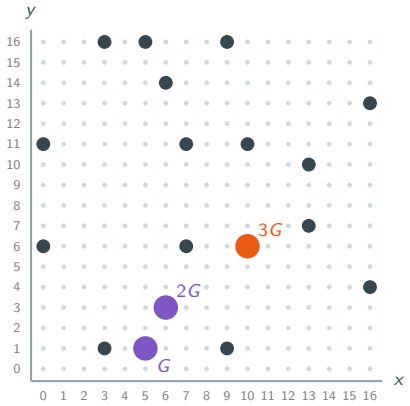
$s = (3 - 1)/(6 - 5) = 2/1 = \mathbf{2}$

$x_3 = 2^2 - 5 - 6 = -7 \bmod 17 = \mathbf{10}$

$y_3 = 2 \times (5 - 10) - 1 = -11 \bmod 17 = \mathbf{6}$

$\mathbf{3G} = (\mathbf{10}, \mathbf{6})$.

Check: $6^2 = 36 = 2 \bmod 17$, $1000 + 20 + 2 = 1022 = 2 \bmod 17$.

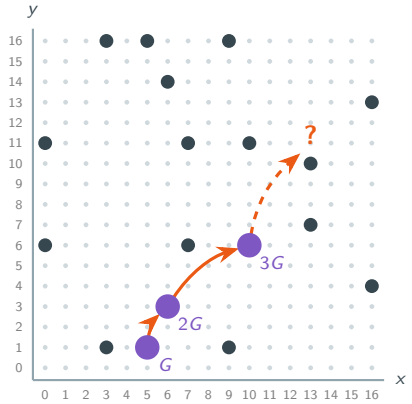


Task 2: all points from G

Task 2
Write the point addition in Python, then add G again and again.

$$G,\ 2G,\ 3G,\ 4G,\ \ldots$$

Predict first
How many different points will you get before you reach O?



Task 2: what to do

Task2/ec_toy.py

```
def add(P, Q):
    if P is 0:
        return Q
    if Q is 0:
        return P
    x1 = P[0]
    y1 = P[1]
    x2 = Q[0]
    y2 = Q[1]
    if x1 == x2 and (y1 + y2) % p == 0:
        return 0
    # TODO 1: the slope s
    #   if P == Q (doubling):   s = (3 * x1^2 + a) / (2 * y1)   mod p
    #   otherwise:              s = (y2 - y1) / (x2 - x1)       mod p
    #   "/ v mod p" means "* pow(v, -1, p)"
    # TODO 2: x3 = s^2 - x1 - x2 mod p,   y3 = s * (x1 - x3) - y1 mod p
    return 0
# TODO 3: print G, 2G, 3G, ... by adding G again and again,
#         until you reach 0. How many points do you get?
```

Task 2

Open `ec_toy.py` from **GitHub**. Write the code in place of the `TODO` lines.

Ready: O (`None`), and the cases $P + O$ and $P + (-P)$.

You write: the slope, x_3, y_3, and the loop.

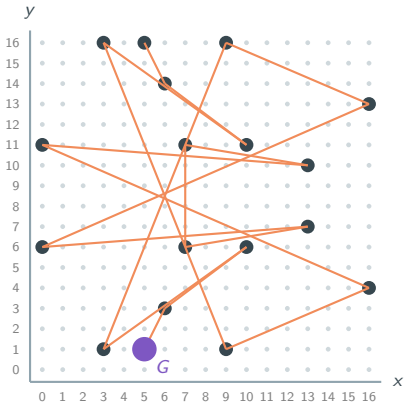
Inverse: `pow(v, -1, p)`

Solution: all points from G

Task2/solution/ec_toy.py

```
if P == Q:
    s = (3 * x1 * x1 + a) * pow(2 * y1, -1, p) % p
else:
    s = (y2 - y1) * pow(x2 - x1, -1, p) % p
x3 = (s * s - x1 - x2) % p
y3 = (s * (x1 - x3) - y1) % p
```

```
G=(5,1)    2G=(6,3)    3G=(10,6)   4G=(3,1)
5G=(9,16)  6G=(16,13)  7G=(0,6)    ...
...        17G=(6,14)  18G=(5,16)  19G=O
```



G visits every point of the curve, then returns to O after 19 steps.

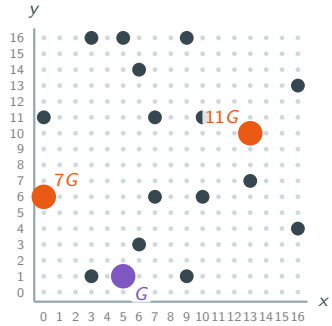
Scalar multiplication

$\mathbf{k}\,\mathbf{G} = \mathbf{G} + \mathbf{G} + \ldots + \mathbf{G}$ (k times). k is a **number**, G and kG are **points**.

$7 \times G =$ (0, 6)

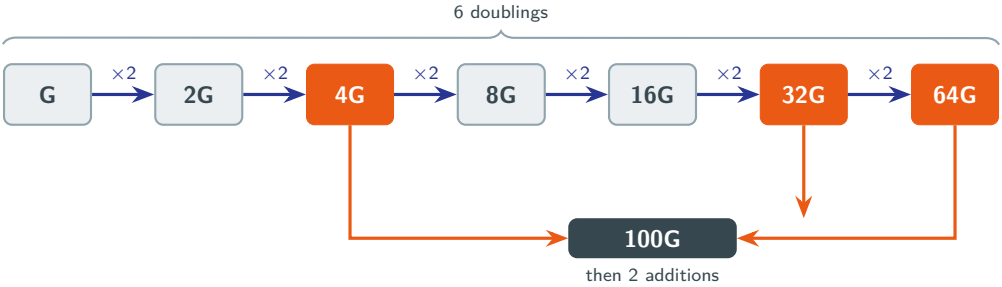
$11 \times G =$ (13, 10)

$19 \times G =$ 0



Fast: double and add

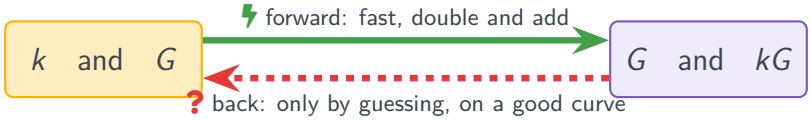
$$100\,G = 64G + 32G + 4G \qquad \text{because } 100 = 64 + 32 + 4$$



For a 256-bit k: about **256 doublings**, not 2^{256} additions.

The hard problem: ECDLP

Elliptic-curve discrete logarithm problem (ECDLP):
given G and $T = k\,G$, find k.



Multiplying a point is easy. Finding the multiplier is hard.

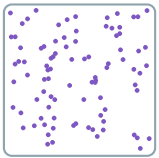
Task 3: find k

Task 3

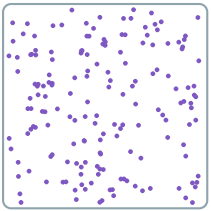
Three curves, p has 16, 20 and 24 bits. You know G and $T = kG$. Find k by adding G again and again.



p: **16 bits**



p: **20 bits**



p: **24 bits**



Predict first

Each p is 4 bits longer. How many times longer will it take?

Task 3: what to do

Task3/ecdlp.py

```python
def find_k(p, a, G, T):
    # TODO: start with R = G and k = 1, keep adding G (R = add(R, G, a, p))
    #       until R == T, then return k
    return None
```

Task 3

Open `ecdlp.py` from **GitHub**. Write `find_k` in place of the `TODO` lines.

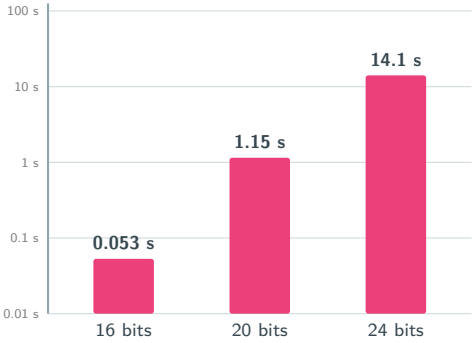
Ready in the file

`add(P, Q, a, p)`, the three curves, and the timing loop.

Solution: find k

Task3/solution/ecdlp.py

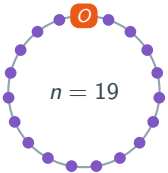
```python
def find_k(p, a, G, T):
    R = G
    k = 1
    while R != T:
        R = add(R, G, a, p)
        k += 1
    return k
```



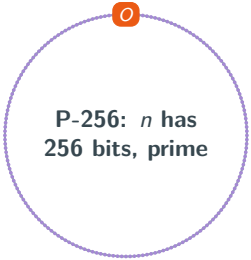
Every 4 more bits make the search more than 10 times longer.

The order of G

The **order** of G: the smallest n with $n\,G = O$.



our curve: only 19 multiples, try them all



about 2^{256} multiples

G must have a huge prime order, or kG takes too few values.

The choice of G matters: a large prime order, or the curve is weak.

Good to know: where do the constants come from?



P-256 (NIST, 2000)

```
b = 5ac635d8 aa3a93e7 b3ebbd55 769886bc
    651d06b0 cc53b0f6 3bce3c3e 27d2604b
```

comes from the hash of a **seed nobody can explain**



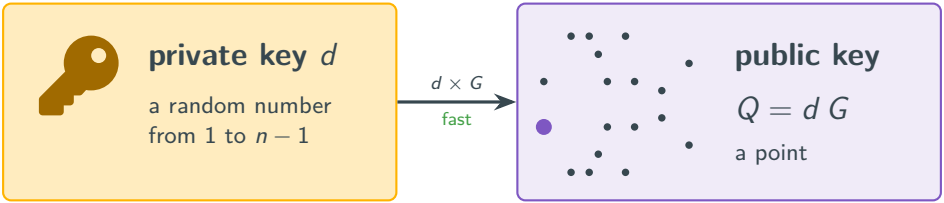
Curve25519 (2006)

$y^2 = x^3 + \mathbf{486662}\,x^2 + x \bmod 2^{255} - 19$

every constant is the **smallest** that meets the stated rules

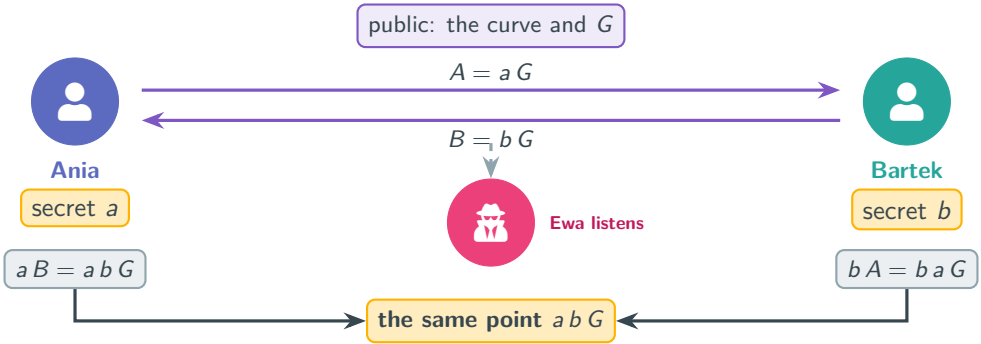
Many prefer curves whose constants anybody can check.

Keys on a curve



The ECDLP protects d: from Q and G, nobody can find it.

ECDH



Both get the same point $a\,b\,G$. Ewa saw only G, A and B.

ECDH is Diffie-Hellman with points instead of powers.

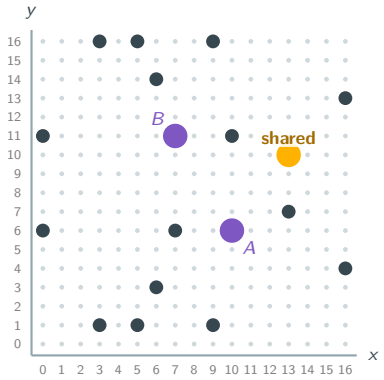
ECDH on our curve

Ania: $a = 3$ $\quad A = 3G = \mathbf{(10,\ 6)}$

Bartek: $b = 10$ $\quad B = 10G = \mathbf{(7,\ 11)}$

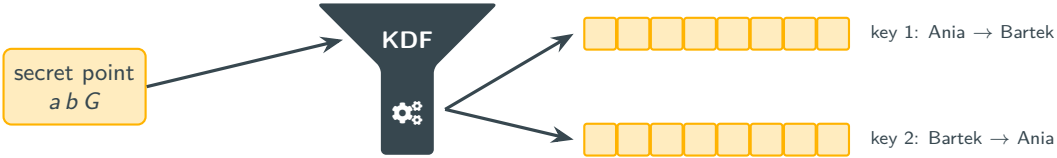
Ania: $\quad 3\,B = 3 \times (7, 11) = \mathbf{(13,\ 10)}$

Bartek: $\quad 10\,A = 10 \times (10, 6) = \mathbf{(13,\ 10)}$



A point is not a key

The shared secret is a **point**. ChaCha20 or AES needs **32 random-looking bytes**.

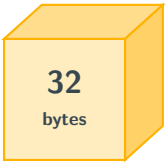


KDF: key derivation function. secret point → one or more symmetric keys.

The common choice is **HKDF** (RFC 5869), built on a hash function.

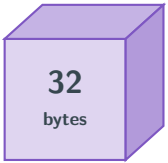
X25519

ECDH on Curve25519, as used on the internet (RFC 7748):



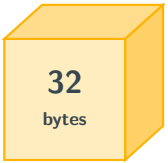
private key

32 random bytes



public key

sent to the other side



shared key

the same on both sides

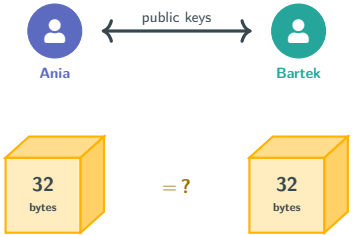
Designed to be hard to implement wrongly.

Task 4: X25519

Task 4
A ready X25519 module in plain Python.

1. Compute Ania's and Bartek's public keys.
2. Compute the shared secret on both sides.
3. Derive a ChaCha20 key with HKDF.

The private keys are the test values from RFC 7748: compare your result.



Task 4: what to do

Task4/ecdh.py

```python
from x25519_ready import public_key, x25519, hkdf

# Private keys: 32 random bytes each (here the test values from RFC 7748, section 6.1)
ania_private = bytes.fromhex("77076d0a7318a57d3c16c17251b26645df4c2f87ebc0992ab177fba51db92c2a")
bartek_private = bytes.fromhex("5dab087e624a8a4b79e17f8b83800ee66f3bb1292618b6fd1c2f8b27ff88e0eb")

# TODO 1: the public keys: public_key(private)            (this is d x G on Curve25519)
# TODO 2: the shared secret on both sides: x25519(my_private, other_public)
# TODO 3: print both secrets (.hex()) and whether they are equal
# TODO 4: derive a 32-byte key for ChaCha20: hkdf(secret, b"lecture 5 chat key")
#         and print it (.hex())
```

Task 4

Open `ecdh.py` from **GitHub**. Write the code in place of the `TODO` lines.

Ready in `x25519_ready.py`

`public_key(private)`, `x25519(private, public)`, `hkdf(secret, info)`

Solution: X25519

Task4/solution/ecdh.py

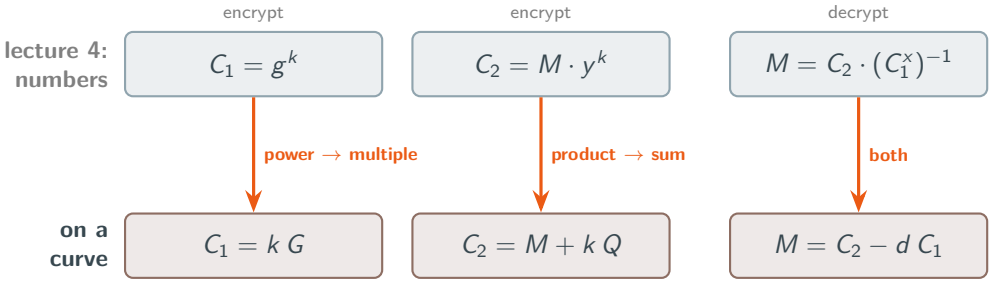
```
ania_public = public_key(ania_private)
bartek_public = public_key(bartek_private)
secret_ania = x25519(ania_private, bartek_public)
secret_bartek = x25519(bartek_private, ania_public)
print("equal:", secret_ania == secret_bartek)
key = hkdf(secret_ania, b"lecture 5 chat key")
```

```
secret:  4a5d9d5ba4ce2de1728e3bf480350f25e07e21c947d19e3376f09b3c1e161742
equal:   True
key:     8232af18b2d158a867dead920cb27b5f70409d17ef4712ba664eb4d2ff392f78
```

The same 32-byte secret on both sides, as in RFC 7748.

ElGamal on a curve

Bartek: private d, public $Q = d\,G$



Encrypt a point M: a random k, $C_1 = k\,G$, $C_2 = M + k\,Q$. Decrypt: $M = C_2 - d\,C_1$.

Exactly ElGamal: powers became multiples, products became sums.

The example on our curve

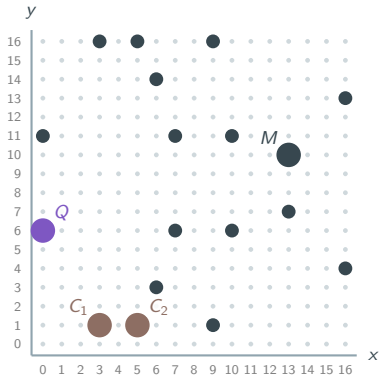
$d = 7, \quad Q = 7G = (0, 6), \quad M = (13, 10), \quad k = 4$

$C_1 \quad = 4G = \mathbf{(3,\ 1)}$

$C_2 \quad = M + 4Q = (13, 10) + (7, 6) = \mathbf{(5,\ 1)}$

decrypt: $d\,C_1 = 7 \times (3, 1) = (7, 6)$

$C_2 - (7, 6) = (\mathbf{13}, \mathbf{10})$ ✔



Task 5: ElGamal on a curve

Task 5
The small curve, Bartek's keys $d = 7$ and $Q = 7G$, the message $M = (13, 10)$.

1. Write `encrypt(M, Q, k)`.
2. Write `decrypt(C1, C2, d)`.

```
def encrypt(M, Q, k):
    # TODO
```

```
def decrypt(C1, C2, d):
    # TODO
```

Ready
`add`, `mul` and `neg` (the point $-P$).

Task 5: what to do

Task5/ec_elgamal.py

```python
def encrypt(M, Q, k):
    # TODO 1: C1 = k G,  C2 = M + k Q.  Return (C1, C2).
    return None


def decrypt(C1, C2, d):
    # TODO 2: S = d C1,  M = C2 - S  (that is: C2 + neg(S)).  Return M.
    return None
```

Task 5

Open `ec_elgamal.py` from **GitHub**. Write the code in place of the `TODO` lines.

Ready in the file

`add(P, Q)`, `mul(k, P)`, `neg(P)`, G, d, Q and M.

Solution: ElGamal on a curve

Task5/solution/ec_elgamal.py

```python
def encrypt(M, Q, k):
    return mul(k, G), add(M, mul(k, Q))


def decrypt(C1, C2, d):
    S = mul(d, C1)
    return add(C2, neg(S))
```

```
$ ciphertext (C1, C2):  ((3, 1), (5, 1))
```

```
$ decrypted:  (13, 10)
```

ECDSA: signing with a curve

private d, public $Q = d\,G$, n = the order of G

sign a message m:

a random k, r = the x-coordinate of $\mathbf{k}\,G \bmod n$

$s = \mathbf{k}^{-1}\,(\mathrm{hash}(m) + r\,\mathbf{d}) \bmod n$ signature (r, s)



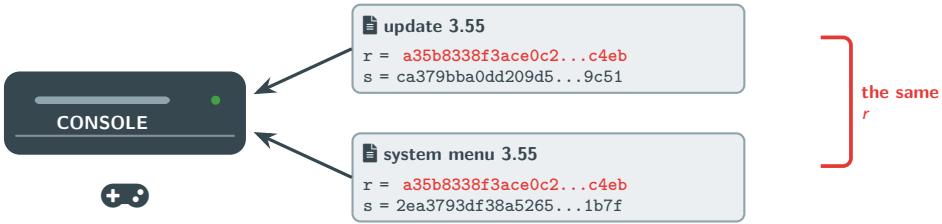
verify: with Q only, checks that r matches



ECDSA: the elliptic-curve digital signature algorithm.

A famous case: the PS3

A games console runs only programs signed with ECDSA by its maker.



2010: researchers (fail0verflow) notice that every signature has the same r.

The same r means the same k. Every time.

Two signatures, one k

$$s_1 = \mathbf{k}^{-1} (h_1 + r\,\mathbf{d})$$
$$s_2 = \mathbf{k}^{-1} (h_2 + r\,\mathbf{d}) \qquad (\mathrm{mod}\, n)$$

subtract: $s_1 - s_2 = k^{-1} (h_1 - h_2) \quad \rightarrow \quad \mathbf{k} = (h_1 - h_2) / (s_1 - s_2)$

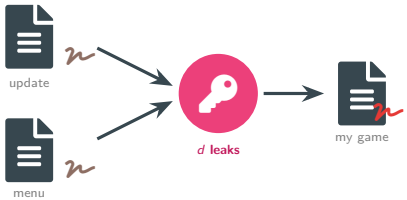
then: $\mathbf{d} = (s_1\, k - h_1) / r$

A repeated k gives away the private key.

Task 6: the same k twice

Task 6
The maker's public key and two signed updates with the same r, on the real curve P-256.

1. Compute k.
2. Compute the private key d and check it against the public key.
3. Sign your own game so that the console accepts it.



Task 6: what to do

Task6/reuse.py

```python
from p256 import G, n, mul, h, sign, verify
# Two signed updates. Look at r: it is the same in both, so the same k was used twice.
m1 = b'Game update 3.55, signed by the console maker'
m2 = b'System menu 3.55, signed by the console maker'
# s = k^-1 (h(m) + r d) mod n.  With the same k and r in both signatures:
# TODO 1: k = (h(m1) - h(m2)) * (inverse of (s1 - s2)) mod n
# TODO 2: d = (s1 * k - h(m1)) * (inverse of r1) mod n
# TODO 3: check that mul(d, G) == Q, then sign your own message with d:
#         sign(b"My homebrew game", d, some new k) and verify it with Q
```

Task 6

Open `reuse.py` from **GitHub**. Write the code in place of the `TODO` lines.

Ready in `p256.py`

`G`, `n`, `mul`, `h` (SHA-256 as a number), `sign`, `verify`.

Inverse mod n: `pow(x, -1, n)`

Solution: the same k twice

Task6/solution/reuse.py

```
k = (h(m1) - h(m2)) * pow(s1 - s2, -1, n) % n
d = (s1 * k - h(m1)) * pow(r1, -1, n) % n
print("private key found:", mul(d, G) == Q)

mine = b"My homebrew game"
r, s = sign(mine, d, 123456789)
print("my game accepted:", verify(mine, r, s, Q))
```

```
$ private key found:  True
```

```
$ my game accepted:  True
```

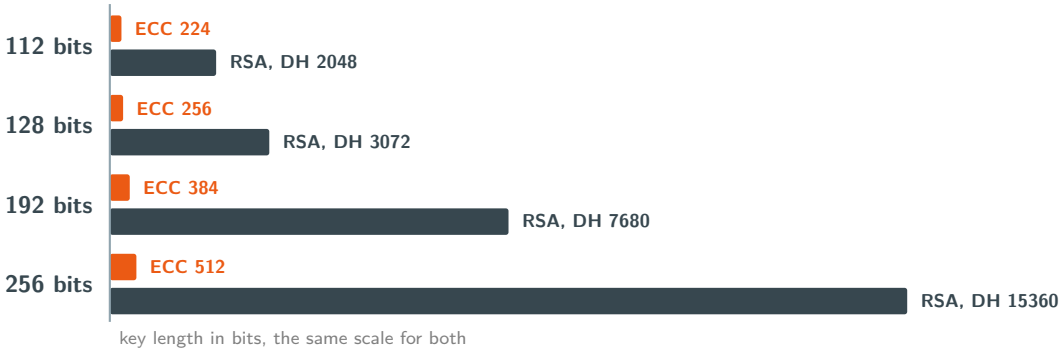
Two signatures with one k, and anybody can sign as the maker.

ECC and ElGamal side by side

	ElGamal (numbers)	**ElGamal on a curve**
operation	multiply mod p	add points
repeated operation	power g^k	multiple $k\,G$
hard problem	discrete logarithm	ECDLP
key for 128 bits	3072 bits	**256 bits**
trap	a repeated k	a repeated k

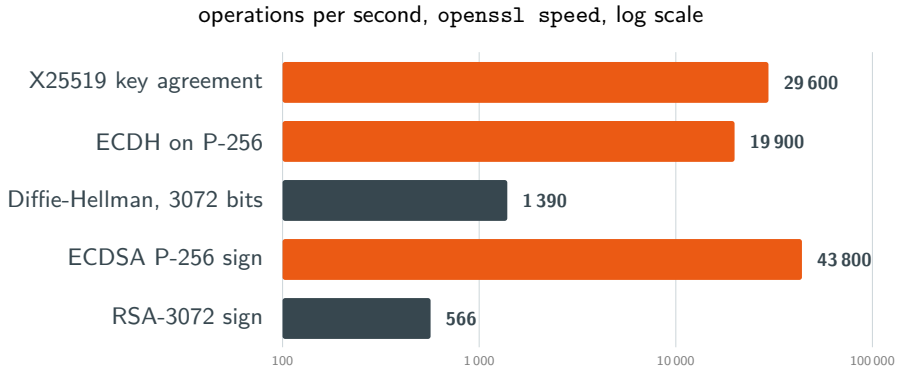
Same algorithm, a different group, keys 12 times shorter.

Key sizes



ECC keys grow slowly. RSA keys grow very fast.

Measured on a laptop



At the same security, ECC is 20 to 80 times faster at private-key work.

Where ECC lives



HTTPS
TLS 1.3 key agreement, mostly X25519; ECDSA certificates



phones
short keys and fast operations save battery



cryptocurrency
Bitcoin signs transactions on the curve secp256k1



messengers
Signal: X25519 for keys, XEdDSA for signatures

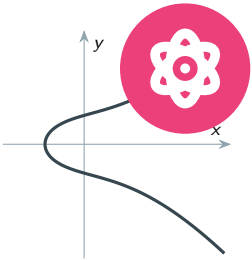


smart cards
small chips with little memory



IoT
sensors and small devices

Good to know: quantum computers



Shor's algorithm
would solve the ECDLP too, on a big quantum computer.

That is why
X25519 is now often paired with a post-quantum algorithm.

Summary

1. ECC gives the same tools as DH, ElGamal and DSA with keys about **12 times shorter**.
2. A curve mod p is a set of points; we **add** them with lines and simple formulas.
3. kG is fast by **double and add**; finding k is the **ECDLP**; G needs a large prime order.
4. **ECDH** (X25519) agrees on a point; a **KDF** such as HKDF turns it into a key.
5. ElGamal and **ECDSA** work on curves too; a **repeated** k gives everything away.